Fig. 1. End-to-End Clinical Thermobiomechanics & DFU Screening Workflow (ThermalSight Framework)

Stage 1: Radiometric Ingestion, Fallback Scaling & Float32 Inpainting

Phase 1A: Raw FLIR Ingestion & Radiometric Matrix Extraction
- Direct extraction of calibrated °C via flyr FLIR unpack
- True physiological temperature matrix [20.0°C–42.0°C]
- Automated parsing of sensor metadata & focal parameters
- High-throughput batch directory queue ingestion

Phase 1B: Adaptive Fallback Scaling & Anomaly Elimination (>100°C)
- Adaptive 8-bit luminance-to-Celsius physiological mapping
- Permanent prevention of raw 0–255 pixel reporting as °C
- Active Python venv resolver ensuring flyr library presence
- Sanitized baseline safeguarding downstream gradient metrics

Phase 1C: OSD Accessory Inpainting & Sub-Degree Float32 Precision
- Bounding-box masking of camera crosshairs & battery OSD
- Telea inpainting maintaining strict float32 decimal precision
- Tissue hotspot protection (lum > 240) preventing dark holes
- Clean plantar background without sub-degree quantization loss

Stage 2: Footprint Morphology Masking & ROI Registration

Phase 2A: Automated 31.0°C Cutoff & Morphological Boundary Smoothing
- Automated 31.0°C thresholding separating ambient floor
- Morphological structuring kernel (7x7) closing pore gaps
- Dual-foot lateral distinction (Right Foot vs. Left Foot)
- Continuous thermal topography ready for gradient sampling

Phase 2B: Strict Binary Footprint Mask & Cold Edge Exclusion (mask_dense)
- Strict binary footprint segmentation isolating skin tissue
- Boundary masking preventing cold background leakage
- Elimination of false cliff gradients on peripheral toe edges
- Normalized 0.5-cell offset indexing for matrix slicing

Phase 2C: Standardized Clinical ROI Registration & Spatial Calibration
- Landmark mapping: Hallux (T1), Metatarsals (M1–M3), Heel
- 36-point boundary vertices for smooth polygonal contours
- Physical distance calibration via spatial scaling (px/cm)
- Session persistence via annotations_session.json

Stage 3: Triad Biomechanics Extraction (PPP Perfusion, PPG Slope & PGA Vector)

Phase 3A: PPP (Plantar Peak Perfusion) & Masked 9x9 Patch Extraction
- Localized absolute perfusion metrics: T_peak, T_mean, T_min, std
- 9x9 ROI patch sampling strictly bounded by mask_dense
- T1 toe min temperature stabilization (34.1°C vs. former 40.5°C)
- Standard deviation normalization capturing focal tissue state

Phase 3B: PPG (Plantar Thermal Gradient) & 2D Sobel Boundary Steepness
- Spatial thermal slope $||\nabla T|| = \sqrt{Gx^2 + Gy^2}$ via Sobel kernels
- Distinction between diffuse physiological heat vs. sharp margins
- Peak gradient (PPG_Peak) identifying acute inflammation summits
- Normalized spatial calibration producing accurate °C/cm gradients

Phase 3C: PGA (Gradient Angle) & Circular Vector Coherence (0–100%)
- Heat dissipation vector $\theta = \arctan2(-Gy, Gx) \bmod 360°$
- Corrected compass: 90° = North (Toes), 270° = South (Heel)
- Circular vector coherence quantifying directional uniformity
- Magnitude-weighted vector averaging preventing 180° flip errors

Stage 4: Multi-Modal Topography, Automated Audit Trail & Clinical Diagnostic Decision Support (Phases 4–5)

Phase 4A: Multi-Modal Topographic Visualization Suite (Fig. 1 A/B)
- 2D Quiver Vector Field mapping heat dissemination trajectories
- 3D Thermal Elevation Surface meshes highlighting hyperemic peaks
- 8-Point Star Radar charts assessing directional steepness (N, NE...)
- High-resolution vector outputs (PNG/SVG/PDF) at 300 DPI

Phase 4B: Automated Audit Trail & Statistical Workbook Synthesis
- Multi-sheet Excel workbook with live formulas (=AVERAGE, =MAX)
- Transparent methodology guide sheet for reproducible peer review
- Comprehensive cohort aggregation (Detailed Long & Wide CSVs)
- Headless CLI non-blocking batch execution for automated labs

Phase 4C: Bilateral Contralateral Asymmetry Screening (ΔT > 2.2°C)
- Contralateral temperature difference screening (ΔT > 2.2°C threshold)
- Differentiation between osteomyelitis/Charcot foot vs. baseline heat
- Identification of sub-clinical tissue stress prior to visible skin breakdown
- Integrated longitudinal risk score for vascular complication tracking

Phase 5: Proactive Clinical Decision Support, Targeted Footwear Offloading & DFU Prevention Policy
- Clinical Decision Support System (CDSS) providing objective thermobiomechanical evidence for multi-disciplinary diabetic foot clinics
- Quantitative biomechanical hotspot mapping directly informing customized orthotic insole fabrication and targeted pressure offloading
- Early-stage pre-ulcerative intervention protocol significantly reducing incidence of Diabetic Foot Ulcers (DFU) and lower-extremity amputations
- Scalable tele-thermology telemedicine architecture enabling continuous remote foot temperature monitoring across home and community clinics